

Pre Incubation Week 4 (DS & Algo) Marathon - Question 3

Create a class **Insurance** with the following attributes

- insuranceId which could be a string value.
- insuranceAmount which could be a float value.
- insuranceType which could be ZERO_DEBT or REGULAR.

Create a class **Vehicle** with the following attributes

- vehicleId of type int or string.
- vehicleRegistration year of type integer.
- vehicleType which could be PRIVATE, COMMERCIAL, SPECIAL_PURPOSE.
- vehicleInsurancePlan which refers to the instance of the associated **Insurance** object.

Create a class **Car** which inherits from Vehicle class and has the following attributes

- carType which could be SUV, SEDAN or HATCHBACK.
- carPrice which could be a float value.
- carColour which could be a string value.

Create a functionalities.cpp file for the following

- A function that checks if all Car instances in a data container passed are of a **carType** value passed as the second argument.
- A function that finds and returns the **vehicleId** of Vehicle with the highest and lowest **insuranceAmount** on the stdout.
- A function that counts and returns the count of Car instances whose **vehicleType** and **vehicleRegistrationYear** matches with values passed as additional arguments to the function.
- A function that accepts a container of Car instances & returns a new container of all Car Instances made up of Car instances whose
 - **insuranceType** is ZERO_DEBT,
 - **vehicleType** is a value passed as an additional argument to the function.
 - **carColour** is a value passed as an additional argument to the function.
- A function that returns a data container with tax values for each Car instance passed in a container to the function. The tax should be 10% of **carPrice** values.

- A function that finds the ***vehicleId*** of the first Car instance from a data container passed argument for which carColour matches a value passed as an additional argument to the function.
- A function that calculates the average carPrice for all instances passed in a data container to the function.
- A function that accepts a container of Car instances and a threshold value for *insuranceAmount* and returns a container of carType values for all Car instances whose insuranceAmount is above the threshold passed as argument.

Demonstrate all functionalities by creating a Main.cpp file.